

P1 Let $G = D_n$ the dihedral group,

$c = \tau$, the transposition, and $a = \sigma$ the rotation.

Then $cac = a^{-1} = a^{n-1}$, or $ac = ca^{n-1}$.

P2. The order of x is $n \Rightarrow$ If $k \equiv l \pmod{n}$, then $x^k = x^l$ in G .

$d = \gcd(k, n) \Rightarrow \alpha k + \beta n = d$ for some $\alpha, \beta \in \mathbb{Z}$.

Assume $\alpha > 0$ or replace (α, β) by $(\alpha+n, \beta-k)$.

Hence $x^d = x^{\alpha k} \in \langle x^k \rangle$. Clearly $\langle x^k \rangle \subseteq \langle x^d \rangle$.

Thus $\langle x^k \rangle = \langle x^d \rangle$. Observe that $x^d, x^{2d}, \dots, x^{n-d}, x^n = e$

are all different and belong to $\langle x^d \rangle$.

In conclusion, $|\langle x^k \rangle| = |\langle x^d \rangle| = n/d$.

P3 If $x^r \in H$, then $x^{r-m} = x^r \cdot (x^m)^{-1} \in H$.

Let $r = km + l$. Then $x^l \in H$ with $l < m$, forcing $l = 0$.

Hence $r = km$, and $H = \langle x^m \rangle$.

By P2, m is a divisor of n , and $H = \{e, x^m, x^{2m}, \dots, x^{n-m}\}$.

P4. $(g^{-1}xg)(g^{-1}yg) = g^{-1}(xy)g$. Thus $(g^{-1}xg)^k = g^{-1}x^k g$.

If the order of x is 3, then $x, x^2 \neq e$ but $x^3 = e$. Then $(g^{-1}xg)^3 = g^{-1}x^3g = e$ for all g .

If $g^{-1}xg$ has order 3, then (taking $g=e$), x has order 3.

Trivial when 3 is replaced by n .